

ABSTRACT

Evolution of Artificial Intelligence in Supply Chain Management: From Analytics to Autonomous Supply Chains

Supply chain management has experienced significant transformation over the years due to globalization, increasing customer expectations, and the growing complexity of business operations. Traditional supply chains relied heavily on manual planning, human expertise, and historical data, which often limited the speed and accuracy of decision-making. As supply networks became larger and more interconnected, organizations began adopting Artificial Intelligence (AI) technologies to improve efficiency, visibility, and resilience across supply chain operations.

The introduction of AI analytics enabled advancements in demand forecasting, inventory optimization, procurement, transportation planning, and risk management. More recently, the emergence of Large Language Models (LLMs) has introduced human-like interaction and the ability to process large volumes of unstructured information within supply chains. However, challenges such as hallucinations and limited access to real-time organizational knowledge highlighted the need for more reliable and explainable AI systems.

Retrieval-Augmented Generation (RAG) has emerged as an effective solution by combining the reasoning capabilities of language models with information retrieved from trusted external knowledge sources. This has led to the development of intelligent enterprise knowledge assistants capable of supporting supply chain professionals in decision-making and problem-solving. This seminar explores the evolution of intelligence in supply chain management, tracing the journey from traditional analytics to AI-powered knowledge systems and discussing the future vision of autonomous, adaptive, and intelligent supply chains.